# Kalahari Ultrasonic Water Meter

LoRa / LoRaWAN | STS Prepaid | Ultrasonic Metering

[Ver1.03A]



## User Manual

Leeroy® SYSTEMS



## Product Introduction

The **Kalahari Ultrasonic Water Meter** is an intelligent smart water meter that integrates ultrasonic metering, STS prepaid management, and LoRaWAN wireless remote communication. The device adopts high-precision ultrasonic measurement technology to collect water consumption data in real time. Through the LoRa wireless communication network, the meter transmits operating status, cumulative water consumption, remaining water credit, alarm information, and other data to the LoRaWAN gateway. The data is then processed by the Network Server and application platform to achieve remote monitoring and intelligent management.

Before delivery, the device has completed basic parameter configuration. Before installation and operation, it is necessary to confirm that the LoRaWAN communication network environment is available, and complete device registration, network authentication, and device binding on the management platform. The unique identification information of the water meter (**Meter Address / DevEUI**) and LoRaWAN network authentication parameters (**JoinEUI, AppKey, etc.**) must be configured and associated in the Network Server and application platform to ensure that the device can successfully join the LoRaWAN network and upload data normally.

After the device installation is completed, the following information should be verified:

- Water Meter Number (**Meter Address**)
- DevEUI (**Device Unique Identifier**)
- JoinEUI (**Application Identifier, if applicable**)
- LoRaWAN network join status
- LoRaWAN frequency band and communication parameter configuration status
- Gateway connection and Network Server configuration status
- Platform device registration and binding status

**MeterAddr:** 0288260510566

## The Kalahari Ultrasonic Water Meter currently supports the following functions:

1. The device adopts ultrasonic measurement technology to achieve high-accuracy flow metering. It can collect and record cumulative volume, instantaneous flow rate, water temperature, battery voltage, valve status, remaining credit (or remaining water volume) and other information in real time. Device data and operating status can be read through the infrared interface.
2. The device complies with the STS Edition 2 international prepayment standard and supports 20-digit standard Token input. Functions include prepayment management, balance control, overdraft management, low balance alarm and automatic valve closure upon credit depletion.
3. The device is equipped with a built-in motorized ball valve. Valve opening and closing operations can be automatically executed according to prepayment status, abnormal alarms or commands issued from the management platform, enabling water consumption control and management.
4. The device adopts LoRaWAN wireless communication technology, communicating with the LoRaWAN gateway through the LoRa wireless link. Data processing and management are performed by the Network Server and application platform. The device can upload data including cumulative flow, instantaneous flow, remaining credit, battery voltage, water temperature, valve status, device operating status, alarm information, and other relevant parameters. Each data transmission includes a timestamp, enabling backend data storage, statistical analysis, and historical data traceability.
5. The device performs hourly data freezing and can store up to 1 year of historical data (expandable to 5 years according to project requirements). Historical data is stored using a circular overwrite mechanism to ensure retention of the latest records.
6. Frozen data and historical records can be read in batches through the infrared interface. (Data format as shown below)

| Time | Total+ | Flow+ | Vol-RTU | Voltage | Water Temp | Status | Consume | Total- | Flow- |
|---|---|---|---|---|---|---|---|---|---|
| 24/09/29 08 | 69.9 | 0 | 3.68 | 0 | 0 | 42 | 0 | 1.2 | 0 |
| 24/09/29 09 | 69.9 | 0 | 3.68 | 0 | 0 | 42 | 0 | 1.2 | 0 |
| 24/09/29 10 | 69.9 | 0 | 3.68 | 0 | 0 | 42 | 0 | 1.2 | 0 |
| 24/09/29 11 | 123.3 | 9.99 | 3.68 | 3.65 | 30.2 | 40 | 53.4 | 1.2 | 0 |
| 24/09/29 12 | 133.1 | 9.99 | 3.68 | 3.65 | 30.2 | 140 | 9.8 | 1.2 | 0 |
| 24/09/29 13 | 143 | 9.99 | 3.68 | 3.65 | 30.2 | 140 | 9.9 | 1.2 | 0 |

7. The device supports leakage detection and burst pipe detection. The operating condition of the pipeline network is analyzed in real time based on historical sampling data. Leakage and burst detection sampling intervals and determination periods can be configured independently to meet project requirements.
8. Device data can be uploaded in packaged mode (up to 24 hours of data per upload, once per day). The upload interval can be configured according to project requirements. Data format shall comply with the "Kalahari Ultrasonic Water Meter Communication Protocol".

| | | | | | | |
|---|---|---|---|---|---|---|
| Hours/Package | 8 | 6 | 4 | 3 | 2 | 1 |
| Reports/Day | 3 | 4 | 6 | 8 | 12 | 24 |

9. The device supports parameter reading and modification through the infrared interface. One-click parameter reading and one-click configuration script writing are supported, facilitating field commissioning and batch configuration.
10. The device supports firmware upgrade through the infrared interface, facilitating future function optimization and product maintenance.
11. The device is powered by a built-in high-energy lithium battery and adopts a low-power-consumption design.
12. Static current consumption of the complete device is less than 55μA, and the service life exceeds 8 years (based on a reporting frequency of once per day).
13. The device supports capacitive touch key triggering for interface switching, status inquiry and data reporting activation, facilitating on-site installation and commissioning. The touch key is designed with anti-misoperation protection to prevent accidental triggering during transportation.
14. The device supports CIU and infrared input keypad communication, enabling Token input, balance inquiry, remaining water credit inquiry, valve status inquiry, and other related operations.
15. The device provides operating status indication, communication status indication and alarm status indication for convenient observation of device operating conditions.
16. The device is equipped with a high-precision Real-Time Clock (RTC).
17. The device incorporates a watchdog management system to prevent abnormal operation and improve system stability.
18. The device adopts an IP68 protection design. Critical internal components are sealed with waterproof technology, making it suitable for long-term outdoor and underground installations.
19. Main technical specifications are as follows:

| Item | Specification | Item | Specification |
|---|---|---|---|
| Product Name | Kalahari Ultrasonic Water Meter | Product Type | Ultrasonic Smart Prepaid Water Meter |
| Prepayment Standard | STS Edition 2 | Communication Method | LoRa + LoRaWAN |
| Nominal Diameter | DN15 | Permanent Flow Rate (Q3) | 2.5 $m^3/h$ |
| Dynamic Range Ratio | R250 | Temperature Class | T50 |
| Pressure Rating | PN16 | Protection Rating | IP68 |
| Electromagnetic Environment Class | E1 | Valve Type | Motorized Ball Valve |
| Data Reading Method | CIU / Infrared Communication | Local Token Input | CIU / Infrared Keypad |
| Remote Token Delivery | Via Management Platform | Overall Weight | Approx. 0.74 kg |
| Battery Replaceable | Yes | | |

## Infrared Token Input Keypad

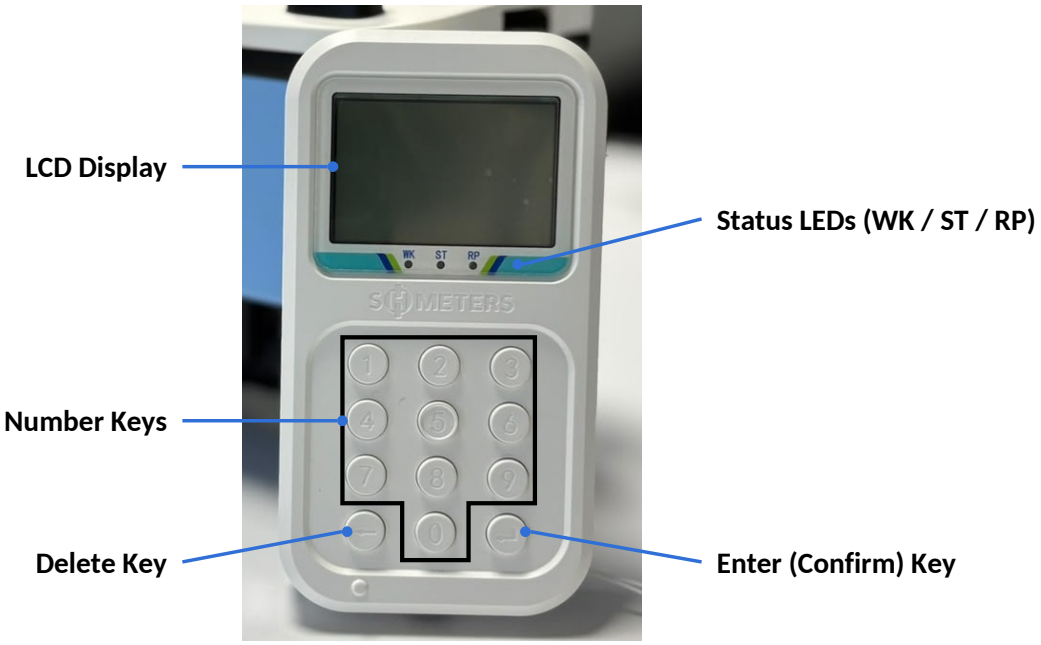
The Infrared Token Input Keypad is a customer operation terminal that communicates with the Kalahari Ultrasonic Water Meter via infrared technology. It is used for Token entry, water meter information inquiry and prepayment-related operations. The keypad provides a convenient user experience, reduces on-site maintenance workload, and improves operational efficiency.

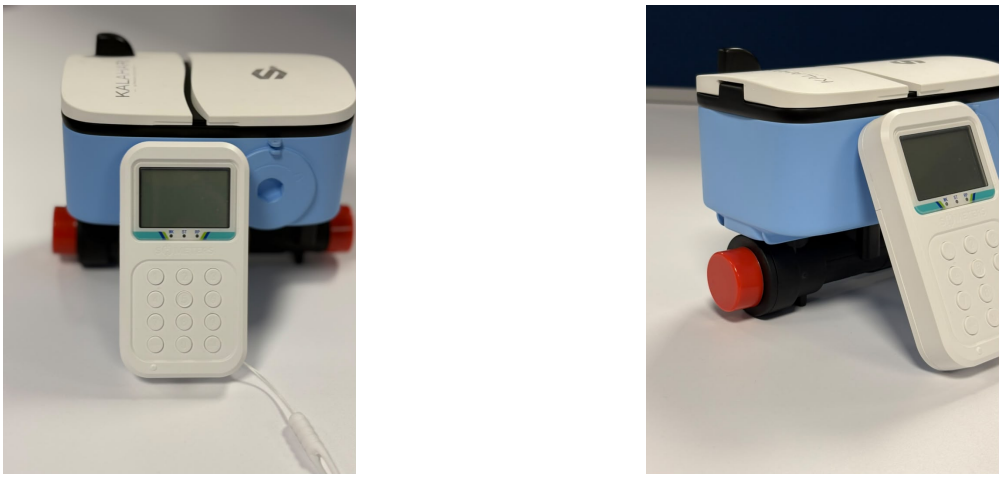
## CIU (Customer Interface Unit)

The CIU is a wireless device used together with LoRaWAN smart water meters to implement STS prepaid recharge and management functions. The CIU device can be powered through a USB interface or by four AAA batteries.

Before use, each CIU must be bound with the corresponding water meter. Only the CIU and water meter with the same address can communicate normally. The binding relationship can be verified by checking whether the Address labels attached to the CIU and water meter are identical.

The CIU device features a clear key layout and simple operation. It consists of 12 keys in total, including numeric keys 0-9, a Delete key, and a Confirm key. A buzzer provides an audible indication after each key operation. The keypad supports input of 2-digit command values and 20-digit STS Token values. The display supports backlight operation for use in low-light environments. Press and hold any key for 3 seconds to activate the backlight.





*Kalahari Ultrasonic Water Meter with its Customer Interface Unit (CIU)*

## 1. Synchronizing Data Between CIU and Water Meter

Press the numeric keys to enter the 2-digit command value "00", then press the Confirm key. The CIU will start synchronizing information with the bound water meter. The display will automatically switch to the "LOAD------" interface and flash periodically.

- If data synchronization is successful, the screen will display "ACCEPT" for 2 seconds, then automatically return to the menu interface and sequentially display the available information.
- If communication with the water meter fails, the CIU will exit the "LOAD------" interface after waiting for 15 seconds and return to the default display interface.

## 2. Recharging the Water Meter Using a 20-Digit Token

Enter the 20-digit STS Token recharge code using the keypad. The display will show the last 8 digits of the entered Token code, and the upper-left corner of the screen will indicate the current number of entered digits. If an incorrect digit is entered, press the Delete key to remove the incorrect input.

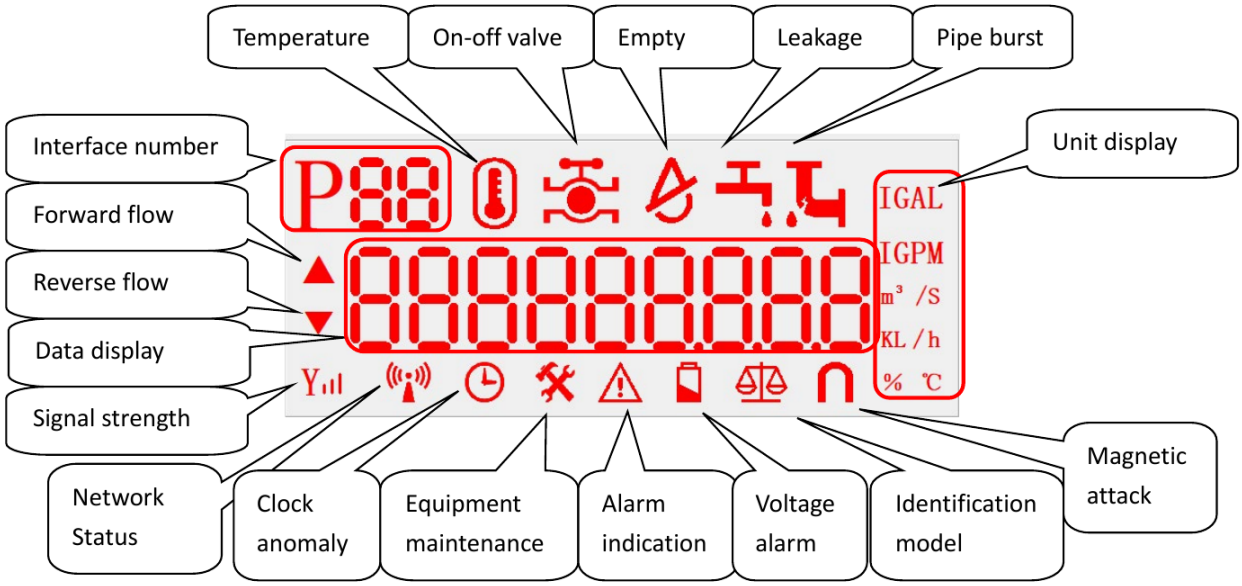
After entering the complete 20-digit Token and pressing the Confirm key, the CIU will automatically execute the Token recharge operation. The display will switch to the "LOAD------" interface and flash periodically.

1. If the recharge is successful, the screen will display "AD $m^3$", indicating the amount of water credit added during this recharge. After a 2-second delay, the CIU will automatically return to the menu interface and display the available information sequentially.
2. If Token verification fails, the screen will display error messages such as:
   a) "CRC ERR" - Token verification error
   b) "TID USE" - Token has already been used
   c) "TID OLD" - Token is expired or invalid
3. If communication with the water meter fails, the CIU will exit the "LOAD------" interface after waiting for 15 seconds and return to the default display interface.

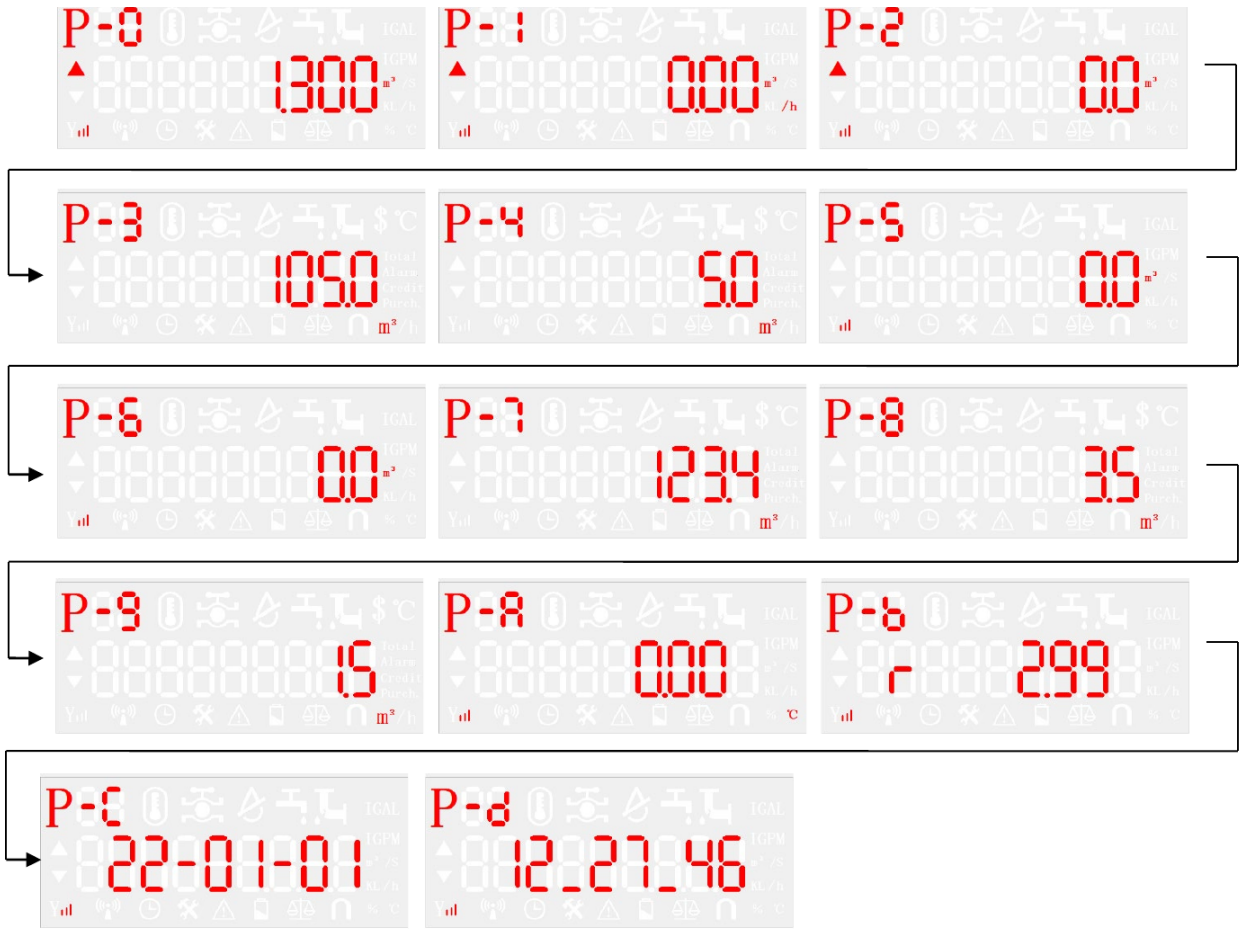
## 3. Manual Menu Switching

When no key value is being entered, press the Delete key to manually switch between CIU display interfaces. Each press of the Delete key will cycle through all available menu pages. If no key operation is detected for a certain period, the CIU will automatically return to the default display interface.

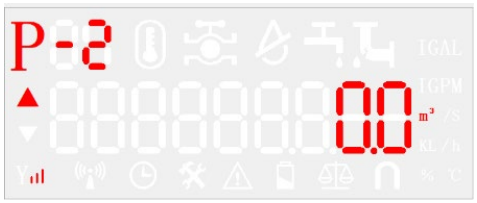
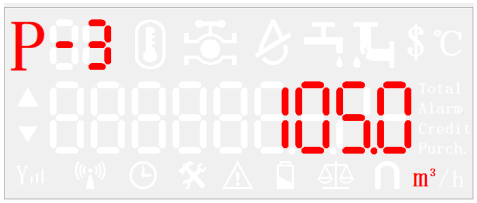
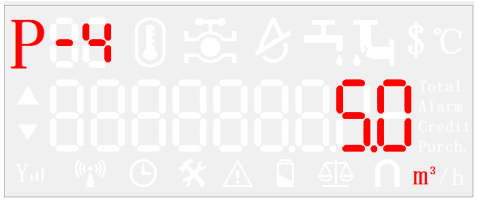
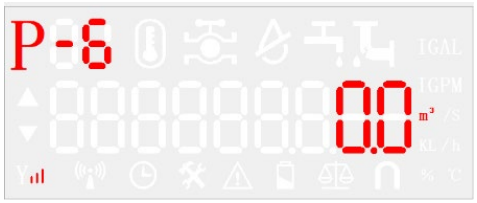
## LCD Display Indications

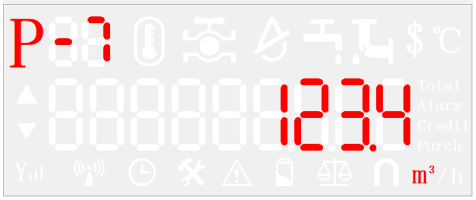
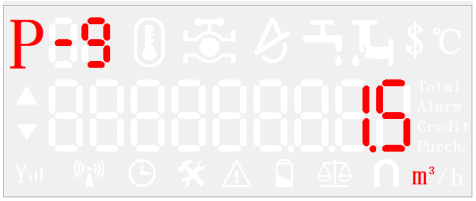
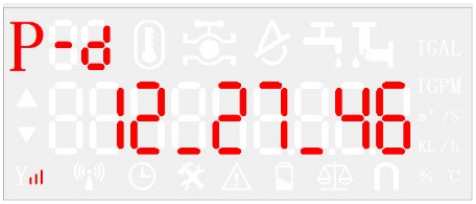



The display interface can be switched using the Quick Trigger Key.



## Interface Description

| Code | Description | Display |
|---|---|---|
| P-0 | Total Meter Reading |  |
| P-1 | Instantaneous Flow Rate |  |
| P-2 | Remaining Water Volume |  |
| P-3 | Total Water Consumption |  |
| P-4 | Total Purchased Water Volume |  |
| P-5 | Latest Recharge Volume |  |
| P-6 | Maximum Recharge Limit |  |

| Code | Description | Display |
|---|---|---|
| P-7 | Accumulated Water Limit |   |
| P-8 | Alarm Water Volume |   |
| P-9 | Overdraft Limit |   |
| P-A | Temperature |   |
| P-B | Battery Voltage |   |
| P-C | Date |   |
| P-D | Time |   |

## Non-Reporting Mode

To minimize power consumption, the water meter is shipped in **Non-Reporting Mode** by default. When the **Non-Reporting Mode icon** appears on the display, it indicates that automatic data reporting is currently disabled.

To restore normal reporting operation, send command **11111** through the infrared interface. After the command is executed successfully, the meter exits Non-Reporting Mode and resumes normal data reporting.

## STS Token Feedback Messages

| Display Content | Description | Display |
|---|---|---|
| AD - 0.5 | Recharge Successful (0.5 $m^3$) |  |
| 1stKCT | First Key Change Token |  |
| 2ndKCT | Second Key Change Token |  |
| 3rdKCT | Third Key Change Token |  |
| Key chag | Encryption Key Changed |  |
| Len err | Invalid Token Length |  |
| Overflow | Exceeds Maximum Water Accumulation Limit |  |
| TID OLD | Old Token |  |

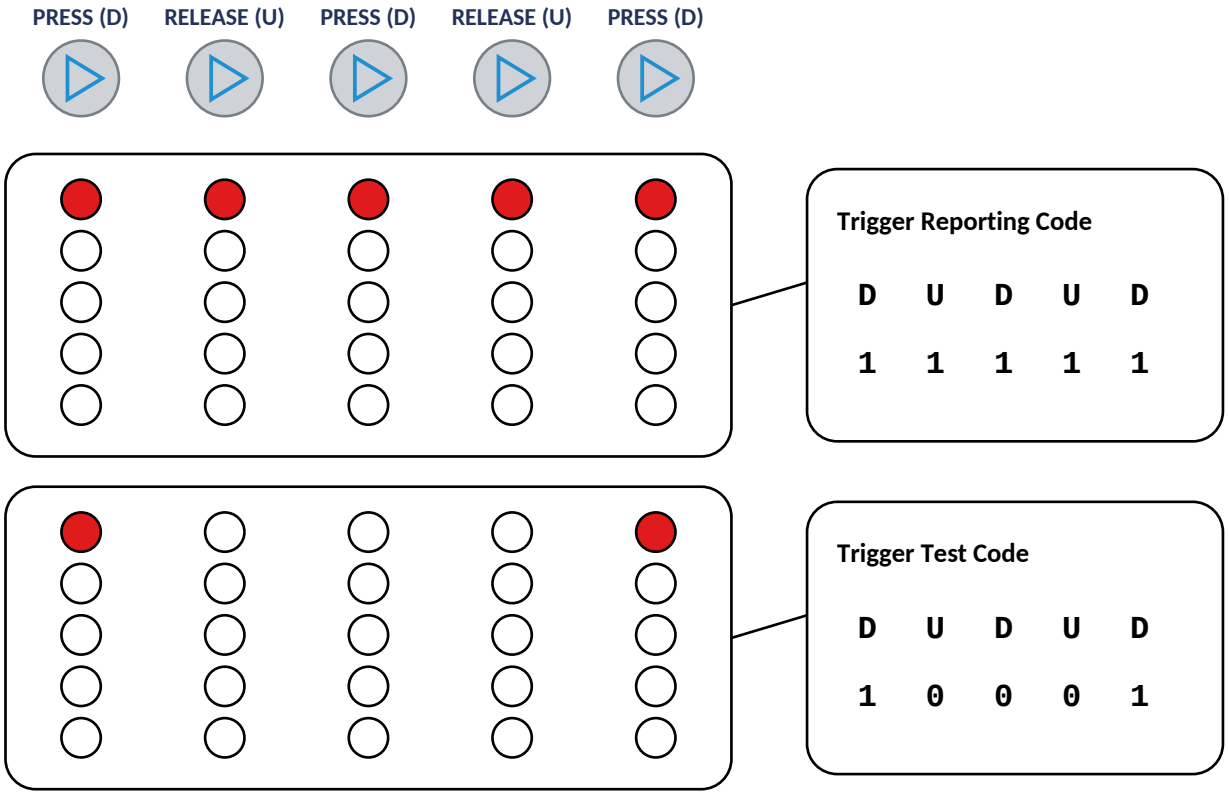
| Display Content | Description | Display |
|---|---|---|
| TID used | Token Already Used |  |
| Accept | Accepted |  |
| Reject | Rejected |  |
| -0.4 | Overdraft / Negative Balance |  |

## Touch Key Operation Codes

```
//*******************************************************
// Key Operation Format <D.U.D.U.D>
// D: Key Press Duration
//    LED flashes once every second (100 ms duration)
//    Value increments by 1 every second, maximum value 5
// U: Key Release Duration
//    LED flashes once every second (50 ms duration)
//    Value increments by 1 every second, maximum value 5
// Two rapid LED flashes indicate command acceptance.
// When triggering the module online command:
//    Blue LED ON = Network registration in progress
//    Red LED flashes three times = Gateway response received
// If a complete command is not entered within 5 seconds,
// the input is considered invalid and ignored.
//*******************************************************
```

| Key Code | Function |
|---|---|
| 10001 | Test Command |
| 11111 | Trigger Data Reporting |

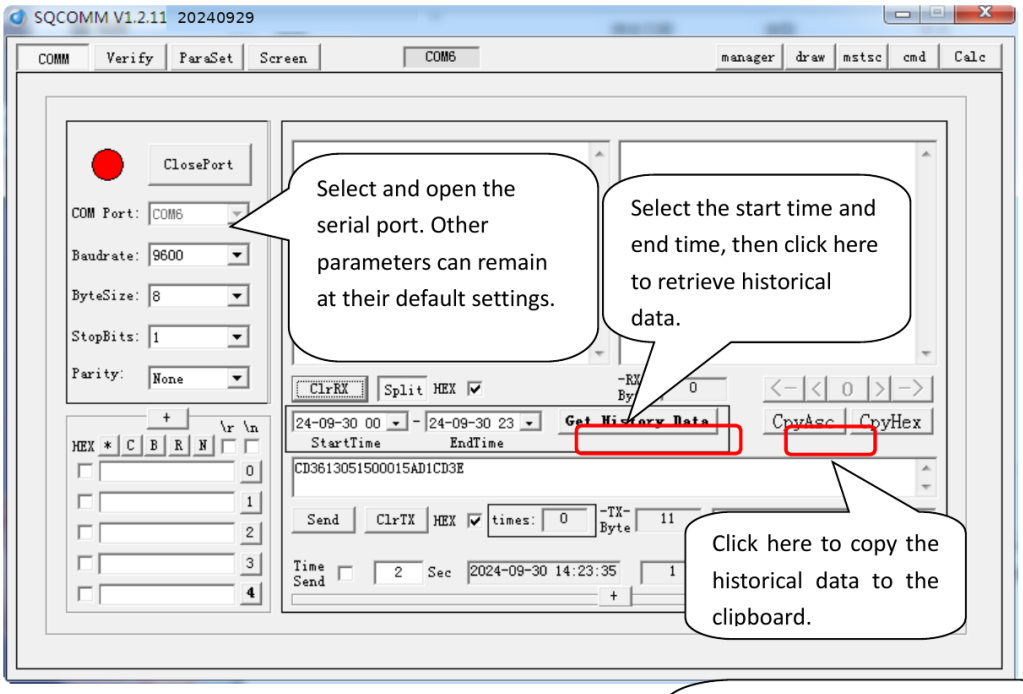
The key trigger code operation procedure is shown below:

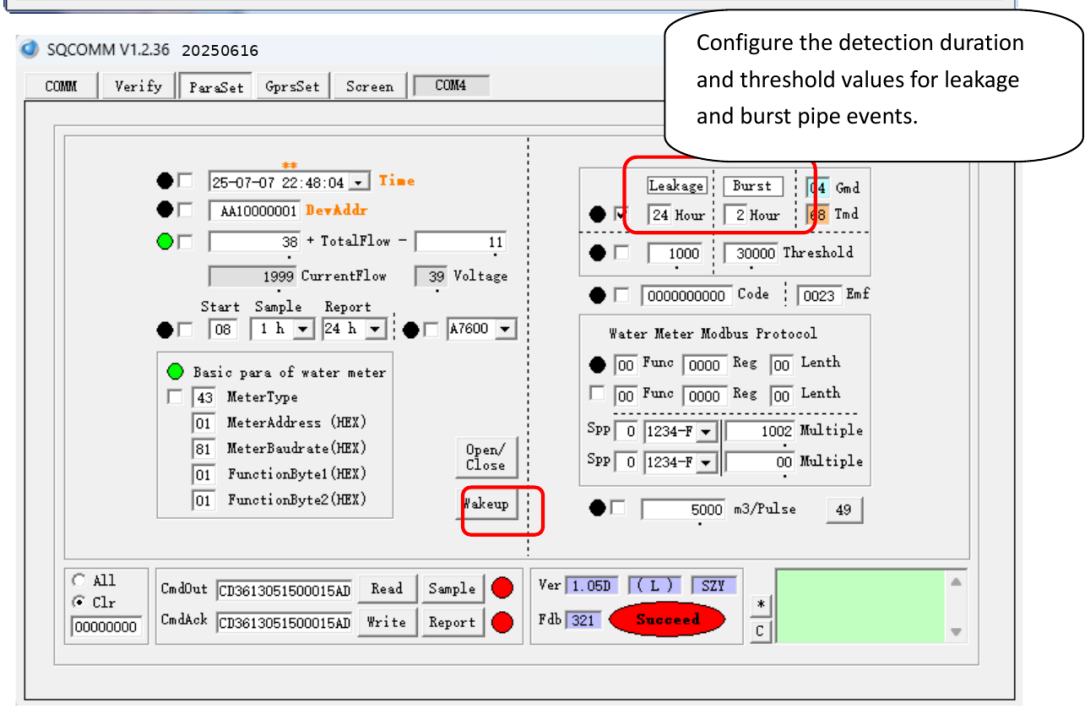



● Red dots indicate the number of LED flashes. Each press or release step can count up to a maximum of 5 flashes; if a step exceeds 5 flashes, the code becomes invalid. To cancel an incomplete code, wait 5 seconds (i.e. 5 flashes). Once the automatic flashing has finished, a new code can be entered.

## Serial Port Software Operation

Open the serial port software.




### Enter the "ParaSet" Parameter Setting Page

Use the Open/Close button to open or close the serial port.
Click the Wakeup button to wake up the device for reading or writing operations.

**Note:** If no operation is performed for 90 seconds, the device must be awakened again.

## Read Parameters

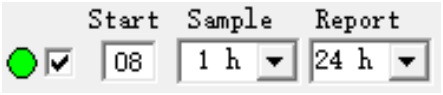
Click the **Read** button to read device parameters. When the reading indicator turns red, the parameters have been read successfully. If the Leakage or Burst Pipe indicator turns yellow, the corresponding event has been detected.

Leakage Leakage Indicator  Burst Burst Pipe Indicator

## Write Parameters

Click the **Write** button to write modified parameters. When modifying parameters, the check box on the left side of the corresponding parameter must be selected. After successful writing, the corresponding indicator lights up to indicate that the operation has been completed successfully.

- **Sample** - triggers immediate sampling.
- **Report** - triggers immediate data reporting.




## Default Settings

1. Sampling Interval: **1 hour**
2. Reporting Interval: **24 hours**

These parameters can be modified as required.

## Common STS Terminology

| Abbreviation | Description | Abbreviation | Description |
|---|---|---|---|
| SGC | Supply Group Code | KMC | Key Management Centre |
| KCT | Key Change Token | KMI | Key Management Infrastructure |
| KEK | Key Exchange Key | KMS | Key Management System |
| KEN | Key Expiry Number | STA | Standard Transfer Algorithm |
| KRN | Key Revision Number | TID | Token Identifier |
| KT | Key Type | POS | Point of Sale |
| EA | Encryption Algorithm | Mfr | Manufacturer |
| TI | Tariff Index | STS | Standard Transfer Specification |
| KUA | Key Use Authorisation Form | IEC | International Electrotechnical Commission |
| HHU | Hand-Held Unit | DRN | Decoder Reference Number (Meter Number) |



*Kalahari Ultrasonic Water Meter*